Morgan Chang - Engineering Physics

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EDUCATION

University of British Columbia | Vancouver, Canada **Sept. 2024-2029 (Expected)**

Bachelor of Applied Science in Engineering Physics

EXPERIENCE

Moment Energy | *Mechanical Engineering Co-op* **Jan 2026 - May 2026**

- Created accurate engineering drawings for the external manufacturing of sheet metal components and hardware for the mass production of a BESS
- Designed and specified components and systems around UL9540, UL1975, NEMA 3R and other regulations to achieve high-voltage and thermal runaway safety certifications
- Designed, performed, and documented 4 in-house preliminary NEMA 3R/4 ingress and corrosion protection tests on 3 unique enclosures
    - Found and fixed 5 ingress points before external certification tests, achieving UL NEMA 3R certification on an enclosure
    - Used results (common ingress areas/failure points) to develop a sealant and gasket installation and inspection procedures, eliminating external test failures

DESIGN TEAM WORK

UBC Thunderbikes | *Aerodynamics Team Lead* **May 2025 -present**

- Designed and executed different manufacturing experiments for creating composite tubing, growing team knowledge on tube manufacturing and composite structural components
- Reviewed CAD designs and manufacturing plans for composite subframe, battery panel, and exterior fairing projects, ensuring overall aerodynamic efficiency and manufacturability

| *Aerodynamics Team Member* **Sept 2024 - May 2025**

- Researched aerodynamic principles and motorcycle-specific studies to design a tail fairing that reduces the drag of the rider-bike system
- Modelled multiple iterations in SolidWorks, which integrated smoothly with the design of the rear frame and gas tank
- Successfully performed FEA and CFD analysis in Ansys, ensuring structural integrity and aerodynamic efficiency
- Planned and executed the wet lay-up technique with extreme attention to detail, decreasing bridging effects and dry spots by about 20% from the previous iteration

PROJECTS

Autonomous Robot (ENPH 253) **Jul. 2025 - Aug. 2025**

- Designed and manufactured a fully autonomous robot for a simulated Mars mission competition
- Electrical system architecture
    - Designed, soldered, and tested the power and boost (motor control) PCBs
    - Tested and debugged H-Bridge (motor control), Schmitt trigger, and IR array PCB's to create a fully functional electrical system composed of 8 (6 unique) PCBs
    - Integrated ESP32 and ESP CAM units for fast and reliable onboard computing
- Mechanical design
    - Designed a robot chassis, drive system, and end effector for 4 unique tasks on the principles of simplicity, reliability, and adjustability
    - Created a detailed 3D CAD model on Onshape, minimizing material waste, iterations, and assembly time
    - Studied and implemented gear systems (claw) and belt drive (drivetrain) to effectively deliver power from our motors
- Software
    - Developed, debugged, and tuned a PID control loop in C++ with sensing from an IR array and IMU
    - Successfully implemented a colour (blob) detection system for an ESP CAM to track model aliens

Go-Kart **Jul. 2025 - Aug. 2025**

- Co-developed the chassis design and manufacturing strategy, optimizing the frame for structural integrity and weight
- Created high-performing custom components using SolidWorks and 3D printing, cutting down project costs and time

TECHNICAL SKILLS

Languages: C, Java, MATLAB, Python

Mechanical: SolidWorks, Inventor, Onshape, Ansys CFD/FEA, CAM, CNC Machining, Laser/Waterjet cutting

Electrical: PCB Design (KiCad), Soldering, PCB testing, High Voltage training